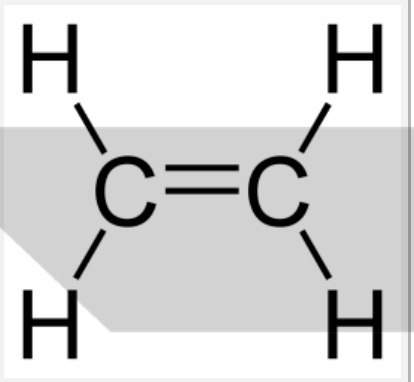
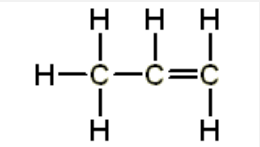
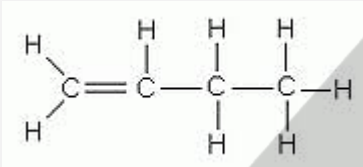
1.4. Alkenes

Alkenes: unsaturated hydrocarbons [at least one double bond between 2 carbon atoms]

Have isomers: same molecular formula but different structural formula (placement of double bond shifts), e.g. but-1-ene and but-2-ene

General formula = C_nH_{2n}

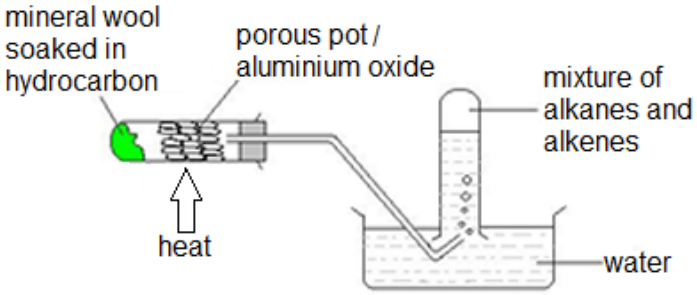
Functional group: C=C bond

(n=1)	Ethene:C_2H_4 (n=2)
N/A	
Propene:C_3H_6 (n=3)	But-1-ene:C_4H_8 (n=4)
	

Catalytic Cracking (Alkane)

- Thermal decomposition reaction at 700 Celsius, in which an alkene (and sometimes hydrogen) is produced from an alkane. It also form shorter-chain alkanes.
- Hydrocarbon is heated, and vapours are passed over a catalyst (alumina (Al_2O_3 and silica)
- **Cracking** always produces a short-chain compound with a C=C bond **(matches the supply of fractions with demand and produces alkene for feedstock.)**

  e.g. Cracking of ethane will give ethene and hydrogen



- Butane → Ethane + Ethene ; $C_4H_{10} \rightarrow C_2H_6 + C_2H_4$

How do we distinguish between saturated and unsaturated hydrocarbons?

Using Bromine water (reddish-brown):

- Saturated: remains reddish brown (unreactive)
- Unsaturated: turns colourless/decolourises

1.7. Alkenes' Addition Reactions

Alkene addition reactions **only form one product.**

1. **With Bromine:** (the test for saturation - orange brown-colourless)

   e.g. ethene (g) + bromine (aq) → 1,2-dibromoethane (l)
2. **With steam (hydration):** forms alcohols with heat (300°C), pressure (**60atm/6000kPa**) and an acid catalyst (phosphoric acid)

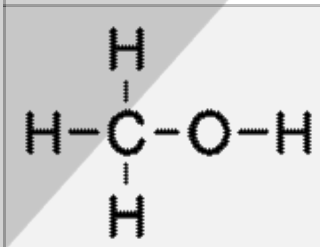
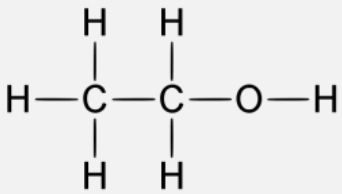
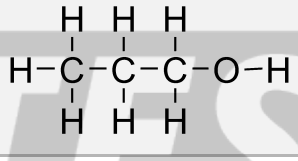
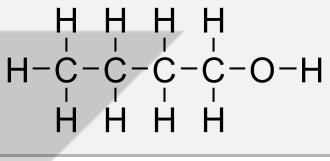
   e.g. ethene (g) + steam (g) ⇌ ethanol (l)
3. **With hydrogen (hydrogenation),** a double bond breaks down to form an alkane with a heat of 200 degrees and a catalyst (nickel)

   e.g. ethene (g) + hydrogen (g) → ethane (g)

1.8. Alcohols

General formula = $C_nH_{2n+1}OH$

Functional group: OH

Methanol: CH_3OH (n=1)	Ethanol: C_2H_5OH (n=2)
	
Propanol: C_3H_7OH (n=3)	Butanol: C_4H_9OH (n=4)
	

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Methods of Production

Fermentation of Aqueous Glucose (for Ethanol)	Catalytic Addition of Steam to Ethene
Yeast is added to dissolved glucose. The products ethanol, carbon dioxide, and the temperature between 25-35 °C are used for optimal enzyme activity.	The ethene reacts with steam (reversibly) to form ethanol in the following conditions: **300°C, 60 atm (6000 kPa).** Catalyst - phosphoric acid [while low temp gives better yield, high temp is used for a faster rate of reaction]
The slow reaction produces a dilute solution that requires processing. Can only be produced in batches	The fast reaction produces pure ethanol. Continuous production (no batches)
Produces greenhouse gas (CO_2)	No greenhouse pollutants
Uses renewable resources	Uses non-renewable resources (crude oil)

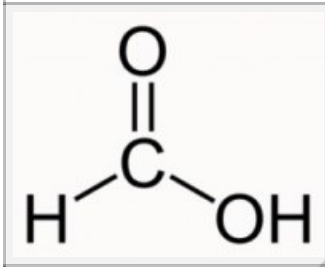
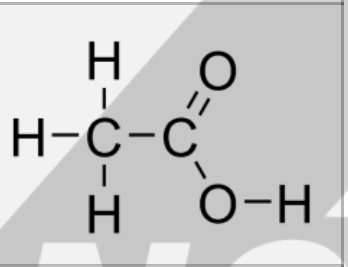
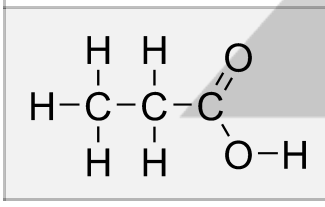
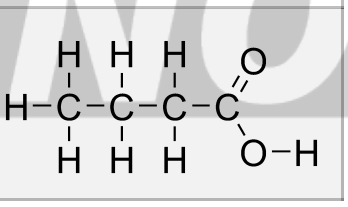
Uses of Ethanol:

- Solvent in glues, printing inks & perfumes
- Fuel

1.9. Carboxylic Acids

General formula: $C_nH_{2n+1}COOH$

Functional group: COOH

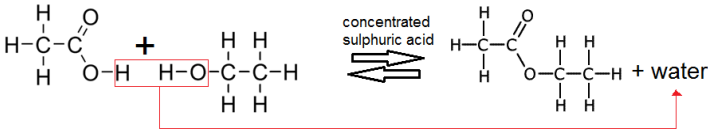
Methanoic Acid: CH_2O_2 (n=0)	Ethanoic Acid: CH_3COOH (n=1)
	
Propanoic Acid: $C_3H_6O_2$ (n=2)	Butanoic acid: $C_4H_8O_2$ (n=3)
	

Properties of Ethanoic Acid

- Weak acid with high pH and low dissociation
- Formed by:
  - Bacterial Oxidation of Ethanol (Vinegar Production)
  - With acidified aqueous potassium manganate (VII) (Good Oxidising Agent)

Carboxylic acids react with alcohols (with an acid catalyst/Heat) to give esters in a **condensation reaction**, for example:

- Ethanoic acid + ethanol ⇌ ethyl ethanoate + water (alcohol = -yl & carboxylic acid = -anoate)



- Carboxylic Acids also have **different structural isomers.**

Reactions with Carboxylic Acid

- Magnesium reacts with ethanoic acid to form magnesium ethanoate and hydrogen gas.
  - $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$
- When propanoic acid reacts with hydroxides, it will form salt and water (Neutralisation reaction)
  - $CH_3CH_2COOH + KOH \rightarrow CH_3CH_2COOK + H_2O$
- When butanoic acid reacts with carbonates, it will form salt, water, and carbon dioxide.
  - $2CH_3CH_2CH_2COOH + K_2CO3 \rightarrow 2CH_3CH_2CH_2COOK + H_2O + CO_2$

1.11. Polymers

- Large molecules are built up from small units known as monomers.
- Monomers are **a single unit of a substance.**
- Polymers can have different linkages depending on the type of polymerisation and monomer

Example:

Small units (monomers)	Linkages
Polyester (PET)	Ester
Polyamide & Protein	Amide

Plastics (PET)

Plastics (**polyethylene terephthalate**) are made from polymers. However, they are not biodegradable. (NOTE: PET can be hydrolysed back to monomers and re-polymerised)

Environment Challenges caused by Plastics

1. Disposal in Landfill sites
2. Accumulation in Oceans
3. Formations of toxic gases from burning

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